7.2. Prove that the Q of the circuit shown in Fig. 7.32(a) is given by Eq. (7.62).

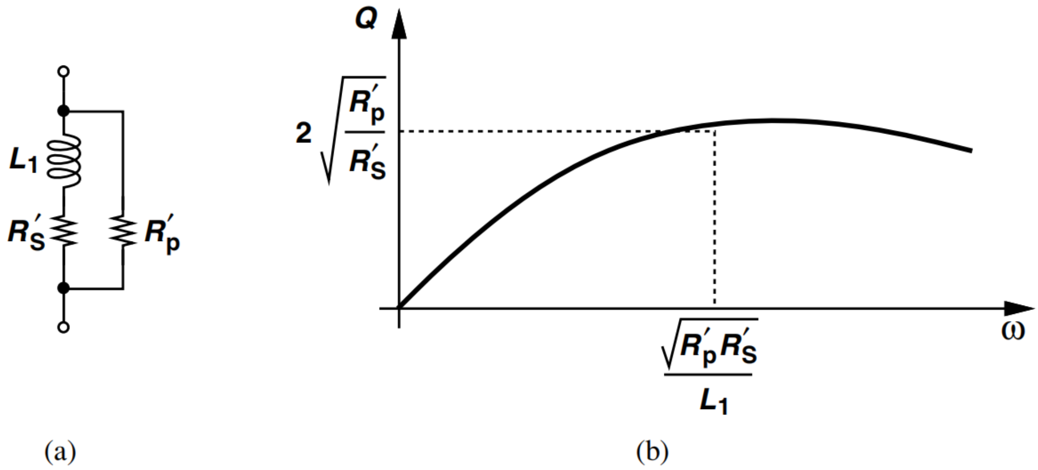


Figure 7.32 *(a) Modeling loss by both series and parallel resistors, (b) resulting Q behavior.*

$$Q = \frac{L_1 \omega R_p'}{L_1^2 \omega^2 + R_S'(R_S' + R_p')}. \tag{7.62}$$

7.4. Using Eq. (7.62), compare the Q's of the circuits shown in Figs. 7.37(b) and (d).

$$Q = \frac{L_1 \omega R_p'}{L_1^2 \omega^2 + R_S'(R_S' + R_p')}. \tag{7.62}$$

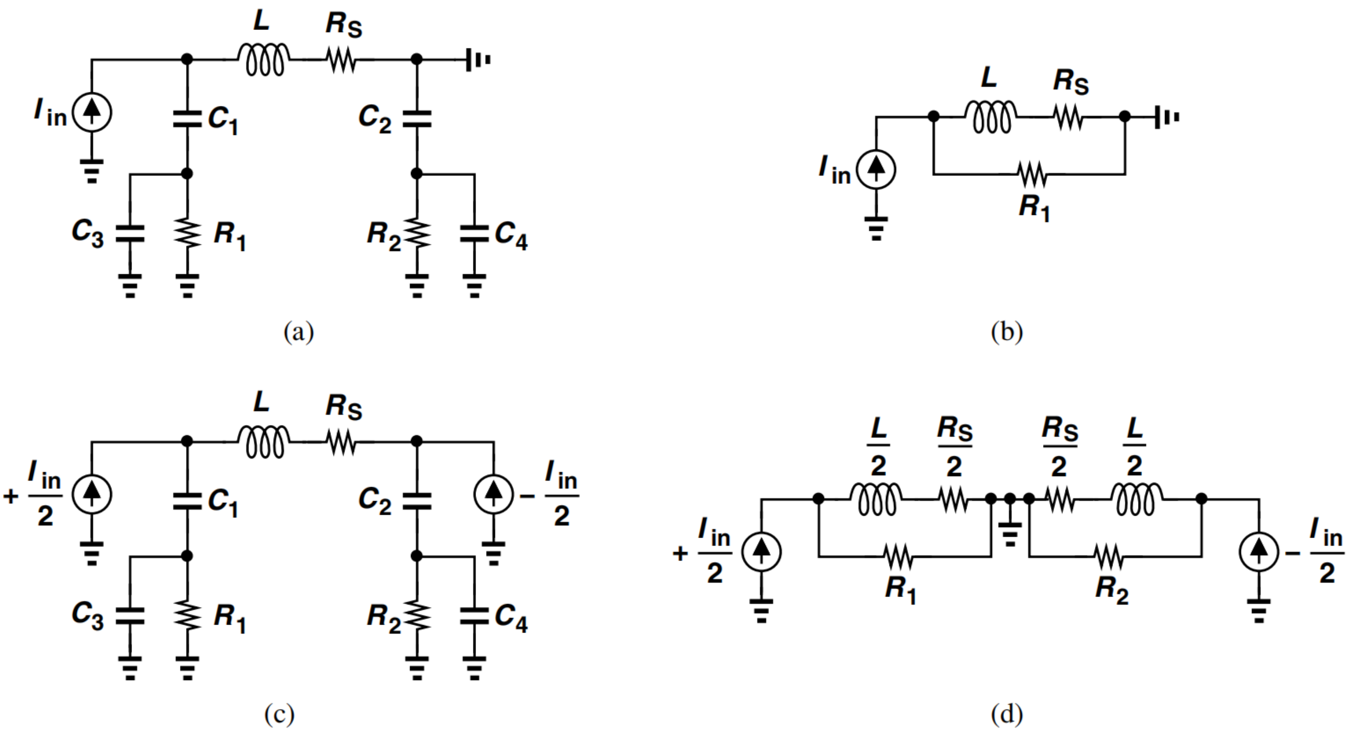
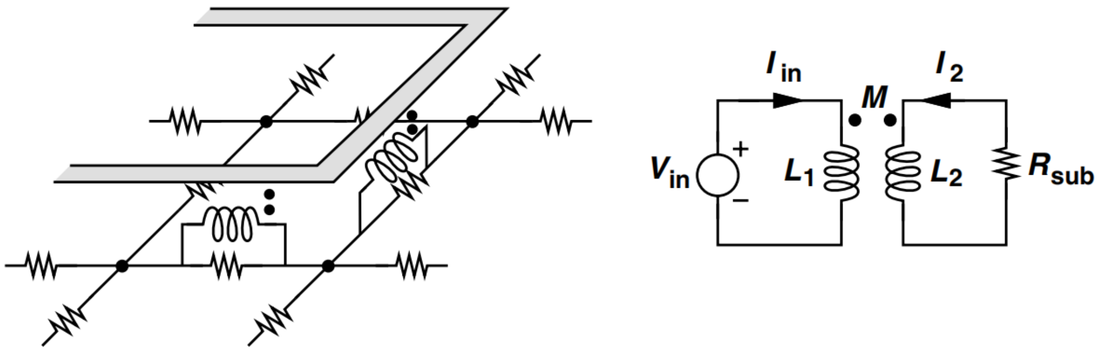


Figure 7.37 *(a) Inductor driven by a single-ended input, (b) simplified model of (a), (c) symmetric inductor driven by differential inputs, (d) simplified model of (c).*

7.3. Prove that for an N-turn spiral inductor, the equivalent interwinding capacitance is given by

$$C_{eq} = \frac{C_1 + \cdots + C_{N^2-1}}{(N^2-1)^2}. \tag{7.125}$$

7.7. For the circuit of Fig. 7.28(a), compute Y_{11} and find the parallel equivalent resistance. Is the result the same as that shown in Eq. (7.55)?



(a) (b)

Figure 7.28 *(a) Modeling of magnetic coupling by transformers, (b) lumped model of (a).*

$$\frac{V_{in}}{I_{in}} = \frac{M^2\omega^2 R_{sub}}{R_{sub}^2 + L_2^2\omega^2} + \left(L_1 - \frac{M^2\omega^2 L_2}{R_{sub}^2 + L_2^2\omega^2}\right) j\omega, \tag{7.55}$$

7.9. Find the input impedance, Z_{in}, in Fig. 7.50.

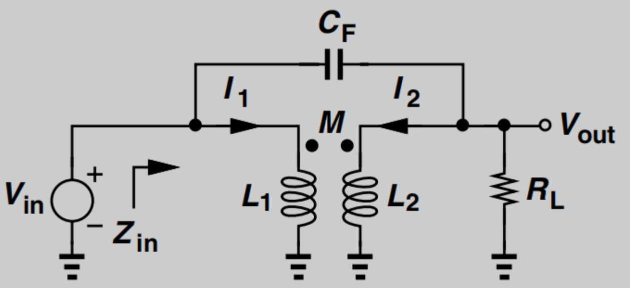


Figure 7.50 *Simple transformer model.*

7.10. Using the capacitance data in Table 7.1, repeat Example 7.25 for an inductor realized as a stack of four metal layers. Assume the inductance is about 3.5 times that of one spiral.

Example 7.25

Compare the equivalent lumped capacitance of single-layer and stacked 4.96-nH inductors studied in Example 7.23. Assume the lower spiral is realized in metal 5 and use the capacitance values shown in Table 7.1.

Table 7.1 *Table of metal capacitances (aF/μm^2).*

	Metal 8	Metal 7	Metal 6	Metal 5	Substrate
Metal 9	52	16	12	9.5	4.4
Metal 8		52	24	16	5.4
Metal 7			88	28	6.1
Metal 6				88	7.1
Metal 5					8.6

Solution:

For a single metal-9 layer, the total area is equal to 2000 μm $\times$ 4 μm = 8000 μm^2, yielding a total capacitance of 35.2 fF to the substrate. As suggested by Eq. (7.26), the equivalent lumped capacitance is 1/3 of this value, 11.73 fF. For the stacked structure, each spiral has an area of 780 μm $\times$ 4 μm = 3120 μm^2. Thus, $C_1 = 29.64$ fF and $C_2 = 26.83$ fF, resulting in

$$C_{eq} = 12.1\,\text{fF}. \tag{7.80}$$

The choice of stacking therefore translates to comparable capacitances.[13] If L_2 is moved down to metal 4 or 3, the capacitance of the stacked structure falls more.